

Approximation of Complex IIR Bandpass Filters Without Arithmetic Symmetry

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Abstract—Complex transfer functions are not restricted to having complex-conjugate symmetry in the frequency-domain, as is the case for real filters. This gives them more flexibility when they are used in communication systems with complex signals, such as intermediate frequency signals of wireless communication systems. This paper describes a set of algorithms and procedures that can be used in solving the approximation problem involved in deriving complex infinite-impulse-response bandpass transfer functions *directly*, without the requirement of first designing a real-transfer-function prototype filter, which is then frequency translated. Because the requirement for a real prototype filter is eliminated, the filters need not have arithmetic symmetry; this results in superior stopbands with smaller filter orders. The procedures can be used for both continuous and discrete-time filters, can allow for arbitrary stopband specifications, and can be used for either equiripple or monotonic passbands.

Index Terms—Complex signal processing, filter approximation, quadrature signal processing, transfer functions.

I. INTRODUCTION

COMPLEX signal-processing blocks are becoming essential components of many communication systems [3]. A complex signal-processing block has two parallel paths with cross-coupling between paths. The topology and gains are identical in both paths; the cross-coupling gains are identical in magnitude but opposite in sign. The signals in one of the paths are normally called in-phase signals; whereas the signals in the other path are usually called quadrature signals. In an ideal complex signal-processing block, the signals in the two paths are in quadrature balance; that is, for every signal component in the in-phase path, there is a corresponding signal component in the quadrature path having equal magnitude and exactly $\pm 90^\circ$ phase difference. The complex operations used are addition, multiplication, and integration (s^{-1}) for continuous-time systems, or the delay operator (z^{-1}) for discrete-time systems. These operations are identical to the operations used for real signals, except that the complex operations are independently applied to both the in-phase and quadrature components. The exception is the multiplication operation, which requires cross coupling. For example, consider multiplication by a fixed coefficient $a + jb$. In this case, the imaginary part of the coefficient (b) is the gain of a cross-coupling path from the in-phase path to the quadrature (or imaginary) path, and $-b$ is the gain of a cross-coupling path from the imaginary path to the in-phase

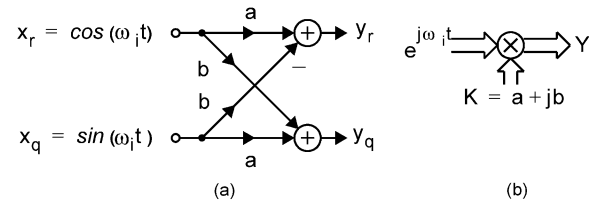


Fig. 1. (a) Real SFG of a complex multiplication. (b) Equivalent complex SFG.

path. The signal flow graphs (SFGs) indicating the multiplication of a single-frequency complexoid by a fixed coefficient are shown in Fig. 1.

This paper discusses the design of complex filters, and focuses on how one obtains complex transfer functions suitable for realizing complex filters. This problem is called the *approximation problem* for complex filters. A related problem is this: given a complex transfer function, how does one obtain a filter realization? This problem is called *synthesis* and is not dealt with in the current paper. Complex filters, which have complex signals and complex coefficients, possess transfer functions that do not necessarily have complex-conjugate symmetry, as compared to real transfer functions which do necessarily have complex-conjugate symmetry. For example, a complex positive-pass filter (PPF) would [4] have a passband at positive frequencies only, and all negative frequencies would be part of the stopband. Filters such as this are often used in the intermediate frequency (IF) stage of a receiver to attenuate images caused by an imperfect preceding quadrature mixer.

Historically, most complex filter transfer functions were obtained by first designing a real lowpass prototype filter, and then transforming this to a complex filter using a frequency transformation. This procedure can be used for both continuous-time filters [1], [5], [17], [29] and for discrete-time and digital filters [17]–[19], [20]–[24]. For example, this is the process used by the function *iirlp2bpc.m*, which is one of the design routines included in the Matlab Filter-Design Toolbox [7], [8]. To the best of the author's knowledge, the design approach based on frequency-translating real prototype filters was the only approach for which programs and utilities were previously available for obtaining infinite-impulse-response (IIR) complex transfer functions, although there has been some publications regarding alternative methods for IIR design such as [9]. There are routines available in Matlab that can be used for the *direct design* of finite-impulse-response (FIR) complex filters, but not for IIR filters, and, in many applications, IIR filters are preferred to FIR filters since they can be more selective for a given filter order.

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Complex transfer functions obtained on the basis of a real lowpass prototype filter necessarily have arithmetic symmetry with respect to a frequency corresponding to dc in the prototype filter. This is a significant limitation in many applications. An alternative approach is to obtain a transfer function directly in the complex frequency domain without first using a real prototype filter. This approach allows for significantly greater flexibility since the transition regions and upper and lower stopbands need not match. For example, a PPF might be designed to have a significantly greater stopband rejection at negative frequencies—where the undesired images generated by the previous nonideal mixer with quadrature imbalance would be present in the input signal—than at positive stopband frequencies above the passband. This design optimizes performance for a given order and complexity.

The design of complex transfer functions without the use of real prototype transfer functions is called the *direct design* of complex filters. The direct design of complex IIR filters with a single passband, but arbitrary stopbands, is the main subject of this paper. The problem of the direct design of complex IIR filters has been dealt with previously in [1] and [2], but, unfortunately, this previous solution was never published in a journal; [1] and [2] are not readily available and do not describe many of the specifics of the algorithms. More important is the fact that the programs and utilities based on these early references are no longer available.¹ For these reasons, and also because the procedures for the direct design of complex IIR filters described here replace and/or significantly extend many of the procedures described in [1] and [2], the current paper is expected to be of value to readers interested in the subject.

The approach taken in this paper is similar to the classical approach for real IIR filters [13], [14], but it is extended to the complex domain using a conformal mapping introduced in [1] and [2]. The approach has other enhancements, such as an extension of the modified-Remez pole-position optimization algorithm, so that both fixed and movable loss poles can be handled. In addition, a new approach to finding stopband minima is described which is robust with arbitrary stopband attenuation specifications that are piece-wise linearly interpolated, and not necessarily constant in a given frequency region. In the proposed approach, the passband attenuation is determined using exact equations. The passband types allowed are equiripple, with the ripple specified, or monotonic, with the attenuation at the passband edges specified. This approach has the advantage that the maximum magnitude loss in the passband is exactly determined to be equal to the specified value. The stopband minima are also equiripple in each region between fixed loss poles. The proposed approach allows for a direct tradeoff between passband loss and stopband margin.

The filter design approach is first presented for continuous-time filters. It is then extended for use in digital filters by using the bilinear transform [25]. It is not necessary (but possible) to have loss poles at infinity for continuous-time filters or, equiv-

¹Those interested in obtaining the complex filter approximation routines described in this paper should send an e-mail request to Ken Martin {martin@eecg.toronto.edu}.

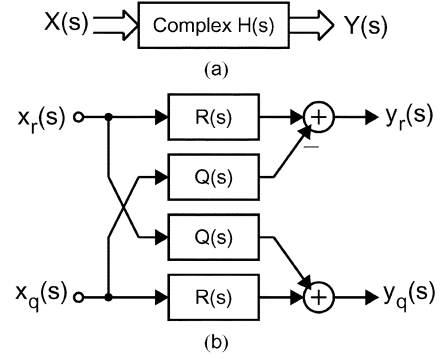


Fig. 2. Realizing a complex filter $H(s) = R(s) + jQ(s)$ by real SFG transfer filters $R(s)$ and $Q(s)$.

alently, at half the sampling frequency ($f_s/2$) for discrete-time filters. At the time of writing this paper, the computing time required on a typical 3-GHz desktop computer is a very reasonable 10–15 s per filter design for high-order filters. This computing time is based on interpreted Matlab [7] routine coding, and would conceivably be less for a compiled implementation; this time saving is not felt to be important. The convergence of the routines has experimentally proven to be robust in most of the examples attempted to date.

II. COMPLEX TRANSFER FUNCTIONS

A real transfer function can be expressed in the form

$$H(s) = k \frac{\prod_{i=1}^m (s + z_i)}{\prod_{i=1}^n (s + p_i)} \quad (1)$$

where s is the Laplace variable, and the coefficients z_i and p_i are equal to the zero and pole frequencies. In addition, the coefficients are either real or occur in complex-conjugate poles. Complex transfer functions are similar to real transfer functions, except that the filter coefficients are not constrained to occur in complex-conjugate pairs, and the input and output signals are possibly complex signals. A complex transfer function can be realized by using four real filters [1], [6], as is shown in Fig. 2. The complex filter $H(s)$ has a transfer function given by

$$H(s) = R(s) + jQ(s) \quad (2)$$

where $R(s)$ and $Q(s)$ are real filters given by

$$R(s) = \frac{H(s) + H^*(s)}{2} \quad (3)$$

$$jQ(s) = \frac{H(s) - H^*(s)}{2}. \quad (4)$$

The notation $H^*(s)$ denotes transfer functions $H(s)$, with all complex coefficients replaced by their complex conjugates. When the filters leading from the real and imaginary branches of Fig. 2 do not exactly match, there are negative-frequency imaging errors (plus less harmful magnitude response errors).

III. CONTINUOUS-TIME COMPLEX BANDPASS FILTER APPROXIMATION

The filter approximation routines described herein are restricted to a single passband, but with arbitrary upper and lower passband frequencies. The only restriction is that the lower

passband frequency is less than the upper passband frequency; both frequencies can be negative, both can be positive, or the lower passband frequency can be negative and the upper passband frequency positive. A low-pass filter can be approximated as a special case where the lower and upper passband frequencies are equal in magnitude, with the former being negative and the latter positive.

The desired stopband rejection is specified in Matlab using two vectors: one that gives the stopband frequencies, and a second vector that gives the desired stopband loss at those frequencies. The attenuation specifications are linearly interpolated between stopband specification frequencies. The closest frequencies to the passband are assumed to be the lower and upper stopband edges. Finally, the specifications outside (below and above) the specified range are interpolated assuming a constant amplitude.

If both upper and lower passband and stopband frequencies are equal in magnitude and of opposite signs, then the resulting transfer functions obtained degenerate into real-transfer-functions. In this degenerate case, the transfer functions obtained are identical to those obtained using classical approximation algorithms [14].

The initial frequency values for movable loss poles are specified in a vector, and the frequencies of all noninfinite fixed-frequency loss poles are also specified in a separate vector. Finally, the number of loss poles at infinity is specified. The pole-placing algorithm used guarantees equal-magnitude stopband minima between each fixed-frequency loss pole, but the margin is normally different in each region between nonmovable loss poles. Since loss poles at infinity are dealt with in a manner identical to other fixed-frequency loss poles, when these are the only unmovable poles specified, then the stopband margins in the upper and lower stopbands will be different. In the case that no fixed-frequency poles (including those at infinity) are specified, then, the margin throughout both upper and lower stopbands will be the same. This is also the case when stopband specifications symmetric about dc are given.

An overview of the complex filter approximation approach described here in consists of the following steps.

- 1) The parameters that need be specified for each filter design are: upper and lower passband frequencies, the passband ripple and type, and lists of stopband frequencies and corresponding attenuation specifications. In addition, the number of loss poles (transmission zeros) at infinity and possibly other fixed frequencies are specified, and initial guesses for movable finite-frequency loss poles that correspond to $j\omega$ axis transmission zeros are given. The frequencies of the finite-frequency loss poles are optimized to obtain equiripple stopband regions during the filter approximation procedure.
- 2) The adaptive loss pole placement (for equiripple stopbands) is done in a transformed domain using a conformal mapping similar to that used previously for real filters [12], where the conformal mapping given by

$$s'^2 = \frac{s^2 + 1}{s^2 + a^2} \quad (5)$$

was used (where the upper passband edge is assumed to be one rad., and the lower passband edge is assumed to be a rad.). However, for complex filter approximation, a different conformal mapping, first introduced in [1], [2], is used. In this approach, the mapping used is

$$s'^2 = \frac{s - j\omega_2}{s - j\omega_1} \quad (6)$$

where ω_1 is the lower passband frequency, and ω_2 is the upper passband frequency. In this algorithm step, all loss poles and attenuation specifications are transformed using (6).

- 3) In the next step, the movable finite-frequency loss poles are adapted to obtain equiripple stopband regions between fixed (unmovable) loss poles. Since $\omega = \infty$ in the s domain transforms to unity on the real axis in the s' domain, then in the case where there are only fixed loss poles at infinity, there will be different attenuation margins for the regions below and above unity in the s' domain. The actual stopband attenuation achieved is dependent on the number and type of loss poles and the passband ripple specified. The update algorithm used is a generalization of the modified-Remez algorithm [10]–[15]. The details of this procedure, which is the essence of this publication, are described in Section VI.
- 4) Obtain the passband reflection zero frequencies by solving exact equations to obtain either an equiripple or monotonic passband.
- 5) Transform the reflection zeros and the loss poles from the s' domain back to the s domain.
- 6) Using Feldtkeller's equation, find the transfer function poles. This involves a root-finding operation.

In the approximation of continuous-time filters, it is assumed all specifications are in radians.

IV. COMPUTER CONSIDERATIONS AND MATLAB CODING

A significant change from the early days of coding filter approximation routines is the Matlab coding environment. This environment provides a wealth of existing routines that can be used, and this minimizes the amount of code that must be written. It is largely an interpreted rather than a compiled environment, but this limitation has not been found to be significant given the current speed of computers. Although much of the Matlab environment can handle the complex math involved in the complex filter approximation problem (this has been the case only recently), not all of it can handle complex transfer functions. For, example, many of the plotting routines currently assume real transfer functions and cannot be used for complex transfer functions. Limitations such as this are being corrected on an ongoing basis. The Matlab Control Toolbox, and, in particular, system objects in this toolbox are used in realizing many of the routines; this is not necessary, but using toolbox objects simplified many procedures. The system objects can handle complex-coefficient transfer functions, although they often give warnings (which are currently suppressed in the distribution supplied by the author).

One of the advantages of using Control Toolbox system objects is that much of the complexity in combining transfer functions, which is often done using root finding, is hidden; complexities such as multiple roots are automatically looked after using robust algorithms that have been finetuned and extensively tested by Mathworks. This paradigm is significant compared to what was previously available when filter approximation routines were first coded in Fortran. The routines generally seem accurate for stopband minima attenuations below around 100 dB, but can suffer numerical inaccuracies at either step four or five (above) for stopband attenuation greater than 100 dB; we hope to address this in the future. The actual order where numerical uncertainty appears is highly dependent on the type of the passband (monotonic is less sensitive), the passband attenuation specification, and the stopband attenuation specification. In general, the routines are typically capable of handling filter orders of 10 to 12 or more without encountering noticeable inaccuracies.²

V. s' DOMAIN FORMULATION

The adaptation of the loss poles (transfer-function transmission-zeros) is done in the s' domain using the conformal mapping given by (6), [1] and [2]. This mapping transforms the transition and stopband imaginary frequencies (in s) to the positive real axis in s' ; infinity is mapped to unity. The upper stopband is mapped to between zero and unity, and the lower stopband is transformed to between unity and infinity. In addition, and perhaps more importantly, the passband is mapped to the complete imaginary axis. This latter mapping allows one to use simple all-pass properties to realize elliptic equiripple passbands exactly [14].

The filter approximation problem posed is a restricted one, where the loss function ($H(s) = 1/(T(s))$, where $T(s)$ is the desired filter transfer function) has the form

$$H(s)H^*(-s) = 1 + \varepsilon^2 K(s)K^*(-s). \quad (7)$$

This formulation is called Feldtkeller's equation [13]. Note that $H^*(-s)$ denotes conjugating all coefficients and then also substituting $-s$ for the Laplace variable s . We constrain the problem for the equiripple passband case³ to have the characteristic function $K(s)$ expressible as

$$K(s) = \frac{F(s)}{P(s)} = \frac{(s - jf_1) \dots (s - jf_{nf})}{(s - jp_1) \dots (s - jp_{np})}. \quad (8)$$

Note that $K^*(-s)$ is equal to $K(s)$ except for a possible sign change.

The parameters $f_1 \dots f_{nf}$ are all real and equal to the passband frequencies where the gain is at a maximum. These are historically called the reflection zeros; therefore, the polynomial $F(s)$ is called the reflection polynomial. The parameters

²A complex filter of order 10, when realized using real circuits, requires a physical filter realization that realizes 20 real poles.

³We will first consider the equiripple passband case only, and then discuss the simpler case of a monotonic passband.

$p_1 \dots p_{np}$ are also all real (possibly zero) and equal to the initial values for the frequencies of the loss poles (both movable and fixed), where the transmission gain is zero. These parameters are therefore often called transmission zeros; they are also commonly called imaginary axis loss poles. The cases of non-imaginary axis loss poles are not dealt with in this formulation. The reflection-zero frequencies are all constrained by

$$\omega_1 < f_i < \omega_2 \quad (9)$$

where ω_1 and ω_2 are the lower and upper passband frequencies respectively. Further, the loss poles are constrained to be either less than ω_1 or greater than ω_2 . (They are further constrained to be outside the transition bands.) Finally, we shall see that the number of reflection zeros nf is given by

$$nf = np + ni \quad (10)$$

where ni is the number of loss poles at infinity. Unless a filter is a high-pass filter, then nf is normally equal to the order of the final resulting filter. In this formulation, it is not necessary that the order of the filter is even, as is required for real bandpass filters. In addition, the formulation can accommodate the case of $ni = 0$, that is, no loss poles at infinity, where the order of the numerator and the denominator are equal. The current formulation does not cover the cases where the order of $F(s)$ is greater than the final filter order. This means that FIR filters are not considered.

Consider the application of the conformal transform defined by (6) to $K(s)$ as defined by (8). Substituting $s = jp_i$ into (6) and noting that p_i is outside the passband range give

$$s'^2 = \frac{p_i - \omega_2}{p_i - \omega_1} = (p'_i)^2 > 0. \quad (11)$$

This transforms $s - jp_i$ to $s'^2 - p_i'^2$ where

$$p'_i = \sqrt{\frac{p_i - \omega_2}{p_i - \omega_1}}. \quad (12)$$

Thus, loss poles on the $j\omega$ axis are transformed to the real axis and the order is doubled. Note also that loss poles at infinity are transformed to unity. Alternatively, substituting $s = jf_i$ into (6) and noting that $\omega_1 < f_i < \omega_2$ give

$$s'^2 = \frac{f_i - \omega_2}{f_i - \omega_1} < 0. \quad (13)$$

This transforms $s - jf_i$ to $s'^2 + f_i'^2$ where

$$f'_i = \sqrt{\frac{\omega_2 - f_i}{f_i - \omega_1}}. \quad (14)$$

Therefore reflection zeros on the $j\omega$ axis are transformed to the imaginary axis, and again, the order is doubled. The resulting characteristic equation is now given by

$$K(s') = \frac{(s'^2 + f_1'^2) \dots (s'^2 + f_{nf}'^2)}{(s'^2 - p_1'^2) \dots (s'^2 - p_{np}'^2)(s'^2 - 1)^{ni}} \quad (15)$$

where again ni is the number of loss poles at infinity.

The filter approximation algorithm now proceeds using the following procedure: first, the movable loss poles p'_i are adapted to equalize the stopband minima, using a generalization of the modified-Remez algorithm and following the formulation described in [13], [14] for unequal stopbands. This generalization is called the *modified-modified Remez (MMR) algorithm*. Next, after each adaptation of the movable loss poles, the reflection zeros f'_i are updated to guarantee an equiripple passband.

At each step of applying the modified-Remez algorithm, it is necessary to find the loss minima of $K(s')K^*(-s')$ in the stopband. In an earlier version, where the stopband attenuation in each region was constrained to be constant between specification frequencies, an adaptive procedure was used to find the minima. However, when the formulation was generalized to allow arbitrary stopband attenuation specifications, with linear interpolation between specification frequencies, the adaptive procedure for finding minima was found to be not robust enough and was dropped in favor of an approach where all attenuations on a fixed grid were searched for the minima. The grid spacing is different for each region between finite loss poles in order to give a prespecified number of points between finite loss poles. This spacing automatically increases the accuracy for regions between closely spaced loss poles and was found to work well with good accuracy, and to still give reasonable computation times. In addition to there being minima between all finite loss poles, minima are also assumed to occur at the stopband edges.

VI. POLE PLACEMENT ROUTINE

The adaptive update used for modifying the loss poles to get equiripple stopband loss is similar to that of [14] with unequal stopbands, except that the procedure has been generalized to allow for a different stopband attenuation margin in each region between fixed loss poles, which are specified a priori. This approach results in a determinate number of equations in the pole-frequency update algorithm. These margins are denoted $M_1 \dots M_{nrngs}$, where $nrngs$ is equal to the number of unique fixed loss poles plus one.

Defining

$$L(s') = \log(K(s')K^*(-s')) \quad (16)$$

then at each iteration, it is necessary to calculate the derivative of the minima $L(s')$ with respect to the individual movable loss poles, p'_i ; that is, we need $DL_{i,j}$, where

$$DL_{i,j} = \left. \frac{d}{dp'_i} L(s') \right|_{s'=\gamma_j} \quad (17)$$

and γ_j is the j 'th minima. Assuming there are nk movable loss poles in the k 'th range, then the number of minima in that range is equal to $nk+1$. The total number of minima is therefore given by

$$n\gamma = np + nrngs = np + nx + 1 \quad (18)$$

where np is the number of movable loss poles and nx is the number of fixed loss poles.

Given this definition, the loss poles are updated by solving the following linear equation at each iteration:

$$\begin{bmatrix} DL_{1,1} \dots DL_{1,np} & -1 \dots 0 \\ \dots & \dots \\ DL_{n1,1} \dots DL_{n1,np} & -1 \dots 0 \\ \dots & \dots \\ DL_{ni,1} \dots DL_{ni,np} & 0 \dots 0 \\ \dots & \dots \\ DL_{n\gamma,1} \dots DL_{n\gamma,np} & 0 \dots -1 \end{bmatrix} \begin{bmatrix} \Delta p_1 \\ \dots \\ \Delta p_{np} \\ M_1 \\ \dots \\ M_{nrngs} \end{bmatrix} = - \begin{bmatrix} A(\gamma_1) \\ \dots \\ A(\gamma_{n1}) \\ \dots \\ A(\gamma_{n\gamma}) \end{bmatrix} \quad (19)$$

where

$$A(\gamma_i) = L(\gamma_i) - \text{Spec}(\gamma_i) \quad (20)$$

and $\text{Spec}(\gamma_i)$ is the specification at the i 'th minimum; that is, for each range there are nk equations of the form

$$[DL_{j,1}\Delta p_1 \dots DL_{j,np}\Delta p_{np}] - M_i = -A(\gamma_j) \quad (21)$$

where the subscript j is iterated over for the number of movable poles in the i 'th range between fixed poles. This update algorithm is an extension of the classical approach, where the number of ranges was constrained to either one, or, in some cases, at most two. For fixed loss poles, the classical approach resulted in an overdetermined set of equations [14]. In the newly proposed algorithm, there is a determined set of equations, since there are $n\gamma$ equations and $np + nrngs = n\gamma$ unknowns. The details of this update are most easily determined through a close examination of the Matlab code in the subroutine `place_poles.m`, which is available from the author.

This update procedure does not move a loss pole from one range, past a fixed loss pole, to an adjacent range; thus, the margin obtained in a particular range is dependent on the number of movable loss poles initially specified in that particular range.

VII. CALCULATING LOG RESPONSE VECTOR AND SENSITIVITY MATRIX

The calculation of the sensitivity of $K(s')K^*(-s')$ with respect to the individual movable poles, at the minima frequency, is one of the most essential operations in the pole placement routine. This operation is considerably simplified by working in the log (base 10) domain. This also improves numerical accuracies. In calculating the sensitivities of $\log\{K(s')K^*(-s')\}$, we ignore the fact that the reflection-zero frequencies are implicitly functions of the pole frequencies (required in order to maintain the ideal passband response). This simplifying approximation has not been found to degrade approximation accuracy. From (15), we have

$$K(s')K^*(-s') = \frac{(s'^2 + f_1'^2) \dots (s'^2 + f_{nf}'^2)}{(p_1'^2 - s'^2) \dots (p_{np}'^2 - s'^2)} \quad (22)$$

TABLE I
ZEROS AND POLES OF EXAMPLES OF FIGS. 3 AND 4

Example of Fig. 3. (k = 8.646576e-005)		Example of Fig. 4. (k = 6.031225e-003)	
zeros	poles	zeros	poles
0 - 1.491795i	-0.036296 + 2.018515i	0 - 1.565412i	-0.113350 + 0.363405i
0 - 0.302354i	-0.108513 + 1.929657i	0 - 0.323422i	-0.363404 + 2.065045i
0 + 0.017085i	-0.171203 + 1.751249i	0 + 0.014896i	-0.622728 + 1.869287i
0 + 2.526607i	-0.198422 + 1.508561i	0 + 2.530644i	-0.740892 + 1.514090i
0 + 2.821155i	-0.175327 + 1.263284i	0 + 2.863571i	-0.630403 + 1.166219i
	-0.115028 + 1.076809i		-0.392030 + 0.954532i
	-0.039358 + 0.979916i		-0.129845 + 0.868304i

and

$$\begin{aligned} & \log\{K(s')K^*(-s')\} \\ &= 2 \{ \log(s'^2 + f_1'^2) + \dots \log(s'^2 + f_{nf}'^2) \\ & \quad - \log(p_1'^2 - s'^2) - \dots - \log(p_{np}'^2 - s'^2) \}. \end{aligned} \quad (23)$$

Note that the special case of poles at infinity in s translates to poles at ± 1 in s' and these do not require special consideration different from the consideration for the movable poles in (22) and (23). Therefore, with the assumption that f_j' is independent of p_i' , for all i, j , we have

$$\frac{\partial}{\partial p_i'} \log\{K(s')K^*(-s')\} = \frac{-4p_i'}{(p_i'^2 - s'^2)}. \quad (24)$$

VIII. CALCULATING CHARACTERISTIC-FUNCTION LOSS-SQUARED MINIMA

At each iteration update, it is necessary to find the minima frequencies of the difference between the squared loss function and the specifications. The specifications are given in the non-transformed domain, and are linearly interpolated between specification frequencies. The reason for not doing the linear interpolation in the transformed domain is that the conformal transform used is nonlinear in frequency; this would result in nonarithmetic symmetry for filters even when the specifications are arithmetically symmetric. At each pole-placement update, the specifications are linearly interpolated at the minima frequencies between specification frequencies, and then the minima found using a grid search algorithm are transformed to the s' domain. They are converted from decibels to log (base 10), and the difference between the log squared loss function and the specifications is calculated.

When finding the minima, a simple and robust but computer intensive approach is used: the margins at a large number of uniformly spaced points between loss poles in the s domain are calculated. An alternative to this simple but computer intensive approach is the use of an adaptive approach, such as one based on the Levenberg–Marquardt optimization procedure [16]. This was used in early versions of the routines, but was found to have problems when the routines were extended to allow for nonconstant specifications with sharp corners. Currently, 200 points between loss points are used. This number results in no noticeable errors, and could be made smaller, but it allows for

the highly accurate approximation of high-order filters in times of only 10–15 s on moderately good home computers.

IX. CALCULATING REFLECTION ZEROS AND FINAL TRANSFER FUNCTION

The calculation of the reflection zeros of the characteristic function is exact; the procedure is the same as described in [14] for real filters. Assume

$$K(s') = \frac{F(s')}{P(s')}. \quad (25)$$

In order to obtain an equiripple passband, one simply takes

$$F(s') = \text{Re}\{P(s')\} = \frac{P(s') + P^*(-s')}{2}. \quad (26)$$

This is implemented in Matlab by first transforming $P(s')$ to transfer function form (as opposed to pole-zero form), so the cancellation of the odd terms implied by (26) is exact. This step is a potential candidate as the major culprit for numerical inaccuracies when stopband minima are greater than 100 dB.

For the case of monotonic passbands, there are $nf = np + nx$ equal reflection zeros where np is the number of movable finite $j\omega$ loss poles, and nx is the number of fixed loss poles (including those at infinity in the s domain). The magnitude of the frequency of the reflection zeros is equal to the *geometric average* of all the loss poles in the s' domain (including those corresponding to infinity in the s domain). Taking into account the fact that there is a repeated loss pole in s' corresponding to each loss pole in s , we have

$$f_0' = \left(\prod_{i=0}^{2nf} p_i' \right)^{\frac{1}{2nf}}. \quad (27)$$

After the reflection zeros are found using either (26) or (27), then, the reflection zeros and loss poles are transformed back to the s domain, and the final loss function, $H(s)$, is found by solving Feldtkeller's equation (that is, (7)) and by choosing the left-half-plane loss-zeros (which correspond to the left-half-plane poles of the transfer function).

X. DISCRETE-TIME FILTERS

Discrete-time digital complex filter approximation is currently supported using the bilinear [25] transform. The filter

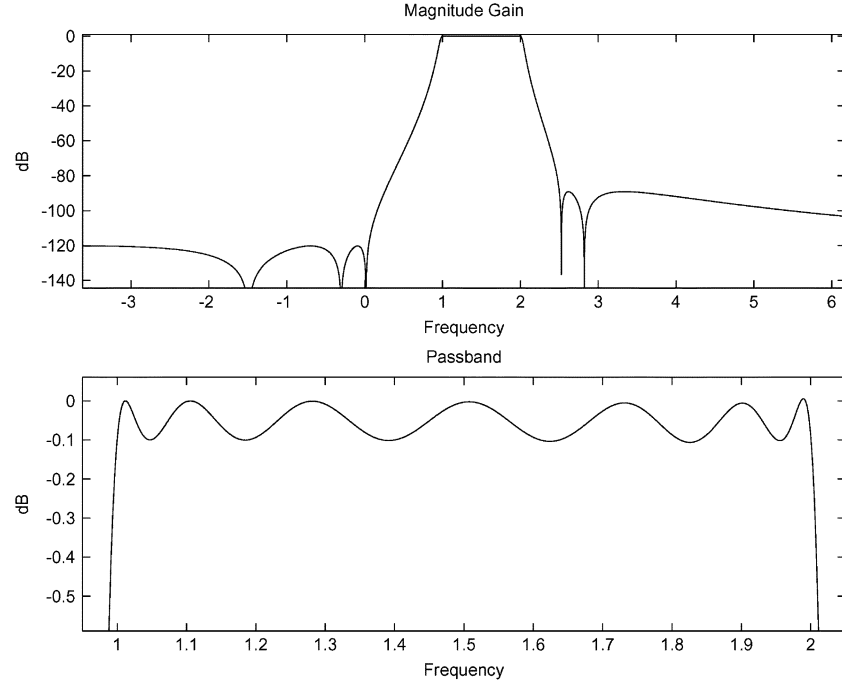


Fig. 3. Magnitude gain and expanded passband response of a continuous-time PPF having an equiripple passband.

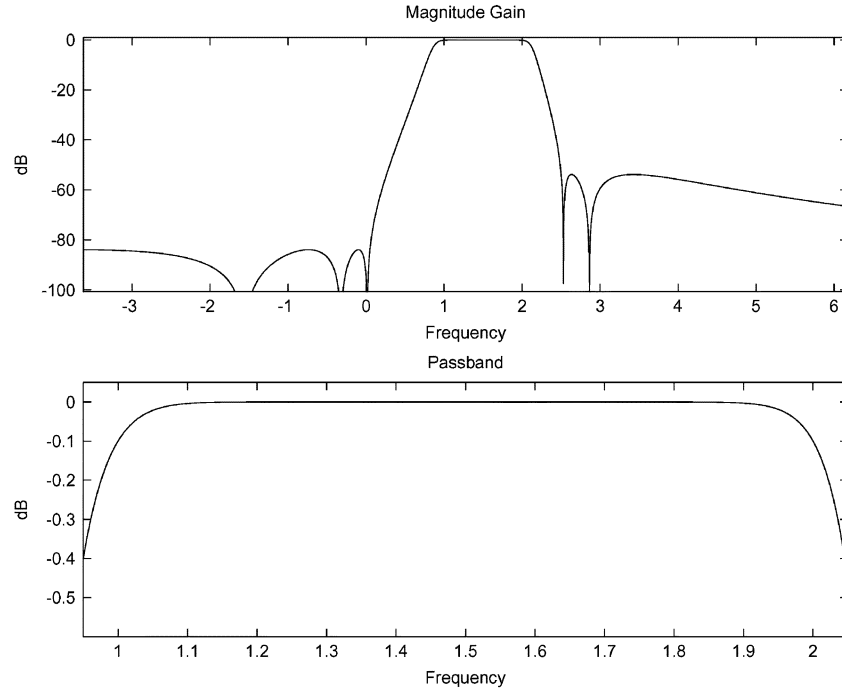


Fig. 4. Magnitude gain and expanded passband response of a continuous-time PPF having a monotonic passband.

specifications of the passband and stopband frequencies (ω_i) are first predistorted using

$$\omega_{di} = 2 \tan\left(\frac{\omega_i}{2}\right) \quad (28)$$

where $-\pi < \omega_i < \pi$ with π equal to half the sampling frequency in radians. After predistortion, a continuous-time com-

plex IIR filter is designed; this filter is then transformed to the discrete-time domain using the bilinear substitution

$$s \rightarrow \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}. \quad (29)$$

This substitution is realized using the Matlab `bilinear()` function operating on the poles and zeros; the control-system toolbox function `d2c(sys, Ts, 'tustin')` was found to not work properly for complex filters at the time this publication was

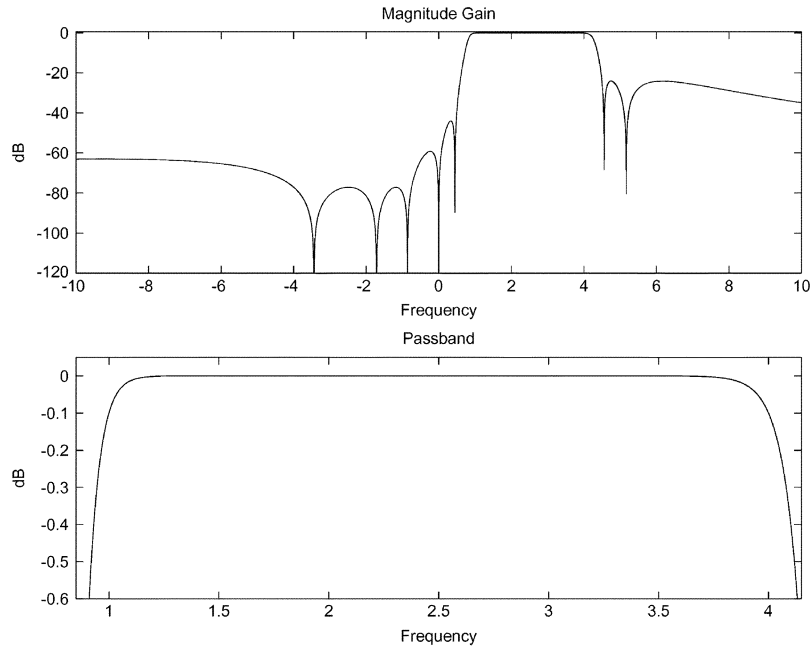


Fig. 5. Magnitude gain and expanded passband response of a continuous-time PPF having a fixed loss pole at dc and additional attenuation at negative image frequencies.

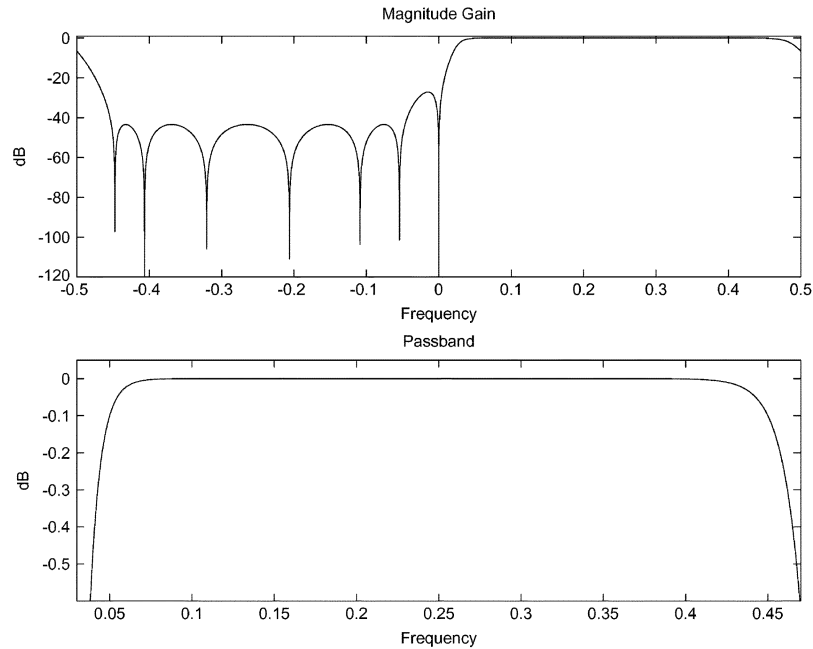


Fig. 6. The magnitude gain and expanded passband response of a discrete-time PPF having a fixed loss pole at dc and additional attenuation at negative image frequencies.

written. For discrete-time filters, it is assumed that all specifications are given relative to the sampling frequency, which is further assumed to be 1 Hz; for other desired sampling frequencies, the filter specification frequencies should first be scaled by dividing them by the sampling frequency.

XI. EXAMPLES

As an example, consider the design of a continuous-time equiripple passband filter having a passband between 1 rad. and 2 rad., a lower stopband at frequencies less than 0.05 rad.,

and an upper stopband at frequencies greater than 2.5 rad. Note that the upper transition region is larger than the lower transition region. This particular filter is an example of a PPF, which has the passband completely at positive frequencies only. In this example, two loss poles at infinity are specified; three movable loss poles in the lower stopband are specified, and two loss poles in the upper stopband are specified. The gain constant, zeros, and poles of the resulting transfer function are shown in Table I. A plot of the transfer-function magnitude loss and an expanded plot of the passband obtained are shown in Fig. 3. Note the accuracy of the passband, even though the

TABLE II
ZEROS AND POLES OF EXAMPLES OF FIGS. 5 AND 6

Example of Fig. 5. ($k = 0.393899$)		Example of Fig. 6. ($k = 0.005570 - 0.069776i$)	
zeros	poles	zeros	poles
0 - 3.440230i	-2.771241 + 2.879010i	-0.945856 - 0.324587i	-0.825230 + 0.047633i
0 - 1.711954i	-1.976151 + 4.234570i	-0.831987 - 0.554794i	-0.520065 - 0.042202i
0 - 0.859803i	-0.795883 + 4.440582i	-0.429044 - 0.903283i	-0.205302 - 0.124060i
0	-0.201639 + 4.281609i	0.271652 - 0.962395i	0.124393 - 0.139026i
0 + 0.448230i	-2.380444 + 1.565190i	0.775789 - 0.630992i	0.427712 - 0.074457i
0 + 4.557535i	-1.609000 + 0.960018i	0.942387 - 0.334525i	0.679747 + 0.034159i
0 + 5.170591i	-0.961808 + 0.789700i	1.000000 + 0.000000i	0.885844 + 0.140086i
	-0.492760 + 0.785198i		
	-0.148843 + 0.814421i		

lower stopband attenuation is greater than 120 dB. The same example, but now with a monotonic passband, is shown in Fig. 4. Note how much larger the stopband attenuation is in the former equiripple passband case.

An example of where one may like to have a noninfinite fixed-frequency loss pole is where a loss pole exactly at d.c. is desired in order to eliminate d.c. offsets that may be present in the input signal of some systems. An example of where one might want to have additional stopband attenuation at some frequencies is a receiver IF where additional attenuation at the negative image frequencies might be desired. Both of these features (d.c. loss pole and additional attenuation at negative frequencies) are illustrated by the filter whose magnitude responses are shown in Fig. 5. This filter was specified to have a passband between 1.0 and 4.0 rad. The desired stopband attenuation was specified at 20 dB everywhere except between -4.0 and -1.0 rad., where 60 dB of attenuation was specified. These frequencies correspond to the image frequencies. Two loss poles at infinity and a fixed loss pole at dc were also specified.

A final example of a wide-band digital PPF filter is given where a passband from $0.05f_s$ to $0.45f_s$ was specified. The desired attenuation at the negative image frequencies ($-0.45f_s$ to $-0.05f_s$) was specified to be 60 dB, and 20 dB elsewhere. No loss poles at infinity (which corresponds to $0.5f_s$) were specified and a single fixed pole at dc was specified. The corresponding magnitude responses are shown in Fig. 6. The gain constants k , zero frequencies z_i , and pole frequencies p_i , as defined by (1), are tabulated in below in Tables I and II. The calling procedures for these and other examples are included in the distribution of the approximation routines, which is available from the author.

XII. CONCLUSION

A set of procedures for designing complex continuous-time and discrete-time filters directly, without first having to design a real prototype filter, have been described. This allows for different attenuation specifications for upper and lower stopbands. Applications where this is beneficial include receivers employing low-IF architectures, for example. The procedures can design filters with either equiripple or monotonic passbands. The procedures can handle any number of mobile loss poles, loss poles at infinity, and fixed-frequency loss poles. This was

made possible by an extension to the modified-Remez algorithm. A number of examples of filter designs with resulting poles and zeros have been provided.

REFERENCES

- [1] W. M. Snelgrove, "Intermediate function synthesis," Ph.D. thesis, Univ. of Toronto, Toronto, ON, Canada, 1982.
- [2] W. M. Snelgrove, A. S. Sedra, G. R. Lang, and P. O. Brackett, (1981) Complex analog filters. [Online]. Available: ftp://ftp.eecg.toronto.edu/pub/software/martin/snelgrove_cmplx.pdf
- [3] K. Martin, "Complex signal processing is not complex," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 51, no. 9, pp. 1823–1826, Sep. 2004.
- [4] M. J. Gingell, "Single-sideband modulation using sequence asymmetric polyphase networks," *Elect. Commun.*, vol. 48, no. 1–2, pp. 21–25, 1973.
- [5] C. Cuypers, N. Y. Voo, M. Teplechuk, and J. I. Sewell, "A general synthesis of complex analogue filters," *Proc. IEE, Circuits Syst., Devices*, to be published.
- [6] E. A. Lee and D. G. Messerschmitt, *Digital Communication*. Norwell, MA: Kluwer, 1988.
- [7] *Matlab, the Language of Technical Computing—Using Matlab*, 2004.
- [8] A. Krukowski and I. Kale, *DSP System Design: Complexity Reduced IIR Filter Implementation for Practical Applications*. Norwell, MA: Kluwer, 2003.
- [9] J. Gryka, I. Kale, and G. D. Cain, "Complex IIR filter design through balanced model reduction of FIR prototypes," *Electron. Lett.*, vol. 31, no. 16, pp. 1332–1334, 1995.
- [10] E. Remez, "Sur le calcul effectif des polynomes de Tchebicheff," *Compt. Rend. Acad. Sci.*, vol. 199, pp. 337–340, 1934.
- [11] B. R. Smith and G. C. Temes, "An iterative approximation procedure for automatic filter synthesis," *IEEE Trans. Circuit Theory*, vol. CT-12, no. 2, pp. 107–122, Mar. 1965.
- [12] H. J. Orchard and G. C. Temes, "Filter design using transformed variables," *IEEE Trans. Circuit Theory*, vol. CT-15, no. 3, pp. 385–405, May 1968.
- [13] G. C. Temes and S. K. Mitra, Eds., *Modern Filter Theory and Design*. New York: Wiley, 1973, ch. 1–3.
- [14] A. S. Sedra and P. O. Brackett, *Filter Theory and Design: Active and Passive*. New York: Matrix, 1978.
- [15] R. K. Henderson, L. Ping, and J. I. Sewell, "Extended Remez algorithms for filter amplitude and group delay approximation," in *Proc. IEE, Circuits, Devices, Syst.*, vol. 138, 1991, pp. 289–300.
- [16] P. Gill, W. Murray, and M. Wright, *Practical Optimization*. New York: Academic Press, 1981, p. 136.
- [17] T. H. Crystal and L. Ehrman, "The design and applications of digital filters with complex coefficients," *IEEE Trans. Audio Electroacoust.*, vol. AU-16, no. 3, pp. 315–320, Sep. 1968.
- [18] R. Boite and H. Leich, "On digital filters with complex coefficients," in *Network and Signal Theory*, J. K. Skwirzynski and J. O. Scanlan, Eds. London, U.K.: Peter Peregrinus, 1973, pp. 344–351.
- [19] M. T. McCallig, "Design of digital FIR filters with complex conjugate pulse responses," *IEEE Trans. Circuits Syst.*, vol. CAS-25, no. 12, pp. 1103–1105, Dec. 1978.
- [20] A. Fettweis, "Principles of complex wave digital filters," *Int. J. Circuit Theory Appl.*, vol. 9, pp. 119–134, 1981.
- [21] X. Chen and T. W. Parks, "Design of FIR filters in the complex domain," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 35, no. 2, pp. 144–153, Feb. 1987.

- [22] P. Regalia, S. Mitra, and J. Fadavi-Ardekani, "Implementation of real coefficient digital filters using complex arithmetic," *IEEE Trans. Circuits Syst.*, vol. CAS-34, no. 4, pp. 345–353, Apr. 1987.
- [23] X. Chen and T. Parks, "Design of IIR filters in the complex domain," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 6, pp. 910–920, Jun. 1990.
- [24] M. Komodromos, S. Russell, and P. T. P. Tang, "Design of FIR filters with complex desired frequency response using a generalized Remez algorithm," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 42, no. 4, pp. 428–435, Apr. 1995.
- [25] A. V. Oppenheim and R. W. Schaffer, *Digital Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1975.
- [26] W. M. Snelgrove and A. S. Sedra, "State-space synthesis of complex analog filters," in *Proc. Eur. Conf. Circuit Theory and Design*, 1981, pp. 420–424.
- [27] G. R. Lang and P. O. Brackett, "Complex analogue filters," in *Proc. Eur. Conf. Circuit Theory and Design*, 1981, pp. 412–419.
- [28] R. Allen, "Complex analog filters obtained from shifted lowpass prototypes," M.A.Sc. thesis, Dept. Elect. Comp. Eng., Univ. of Toronto, Toronto, ON, Canada, 1985.
- [29] A. Sedra, W. Snelgrove, and R. Allen, "Complex analog bandpass filters designed by linearly shifting real low-pass prototypes," in *Proc. Int. Symp. Circuits Systems*, vol. 3, 1985, pp. 1223–1226.
- [30] Q. Liu, "Switched-capacitor complex filters," M.A.Sc. thesis, Dept. Elect. Comp. Eng., Univ. of Toronto, Toronto, ON, Canada, 1985.
- [31] C. Cuyper, N. Voo, M. Teplechuk, and J. Sewell, "The general synthesis of complex analogue filters," in *Proc. Int. Conf. Electronics, Circuits and Systems*, vol. 1, 2002, pp. 153–156.



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